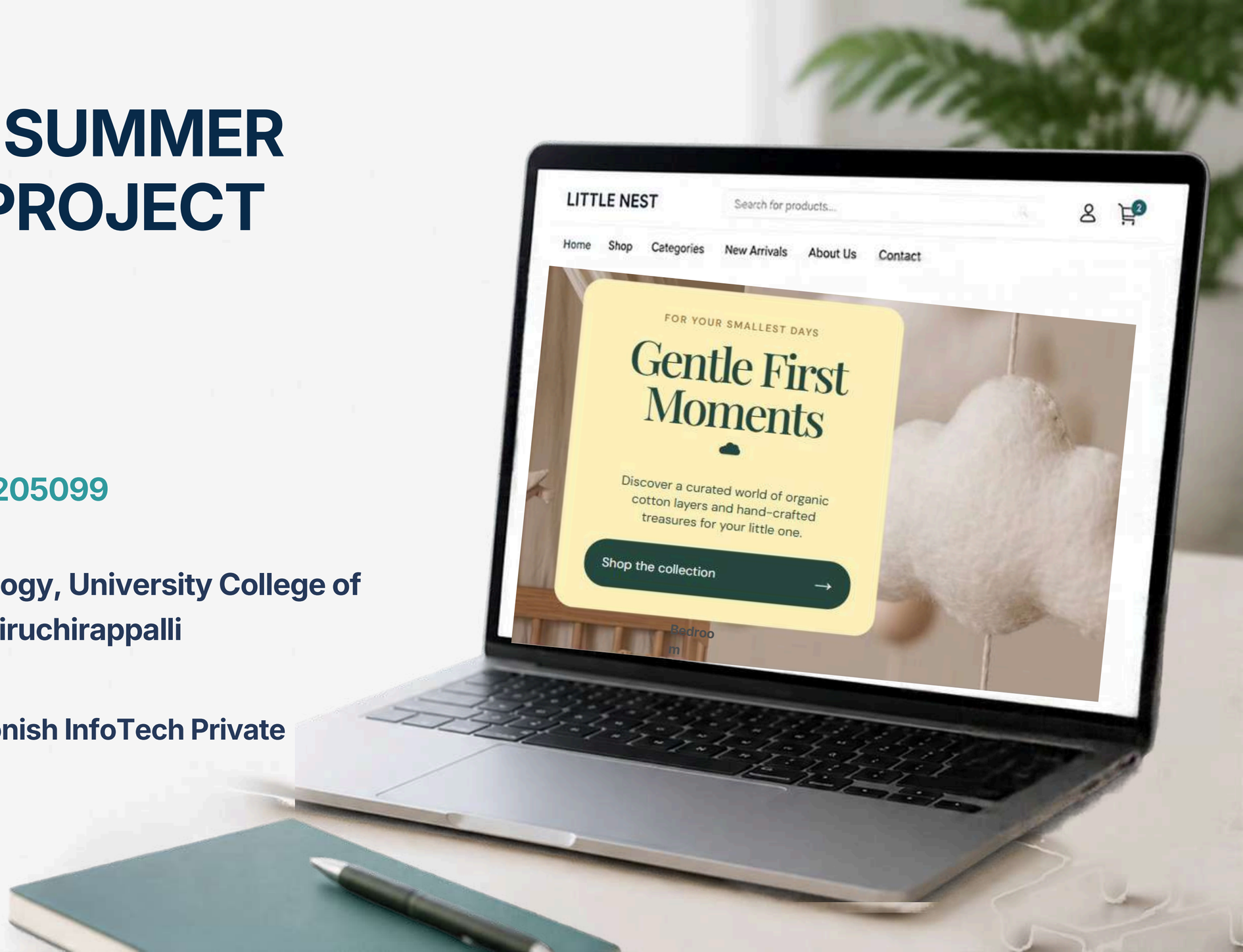
ONE-MONTH SUMMER INTERNSHIP PROJECT REVIEW

Presented by: Harini S.

Register Number: 810023205099

B.Tech Information Technology, University College of Engineering, BIT Campus, Tiruchirappalli

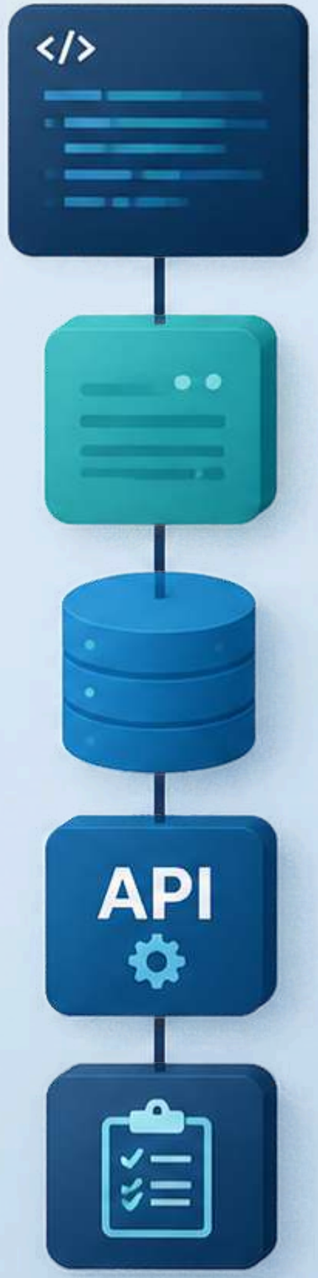
Internship Organization: Astonish InfoTech Private Limited, Tiruchirappalli



Internship Overview

ORGANIZATION	Astonish InfoTech Private Limited
LOCATION	Technology Park, Bharathidasan University Campus, Kajamalai, Tiruchirappalli
INDUSTRY	Information Technology and Software Development
INTERNSHIP AREA	Java Full Stack Development
PROJECT	Little Nest – E-Commerce Website
DURATION	One Month

Practical exposure to **frontend, backend, database integration, REST API development,** and **software testing.**



Project Overview

Little Nest is a full-stack e-commerce web application built for **convenient online shopping.**



DEVELOP RESPONSIVE PLATFORM

Build a responsive online shopping platform for seamless user experience.



USER REGISTRATION AND LOGIN

Enable secure user registration and login functionality.



BROWSE AND SEARCH PRODUCTS

Provide product browsing, search, and detailed information.



CART AND ORDER MANAGEMENT

Enable shopping cart management and order placement.



STORE DATA SECURELY

Store user, product, order, and contact information in MySQL.



INTEGRATE WITH REST APIS

Integrate frontend and backend seamlessly using REST APIs.

Problem Statement and Proposed Solution

EXISTING SYSTEM

- Manual or inefficient order management
- Limited product accessibility and search
- Inadeqate user authentication
- Difficulty managing customer and order data
- Limited online customer communication

PROPOSED SYSTEM

- Centralized digital shopping platform
- Product browsing with secure authentication
- Cart management and order processing
- Customer contact form
- Responsive user interface

Technologies Used

FRONTEND


React.js



HTML



CSS



JavaScript


React Router

BACKEND


Java


Spring Boot


Spring Security


Spring Data JPA

DATABASE



MySQL

API AND INTEGRATION


RESTful APIs



Axios

DEVELOPMENT TOOLS


Visual Studio Code



IntelliJ IDEA


Apache Maven



Node.js

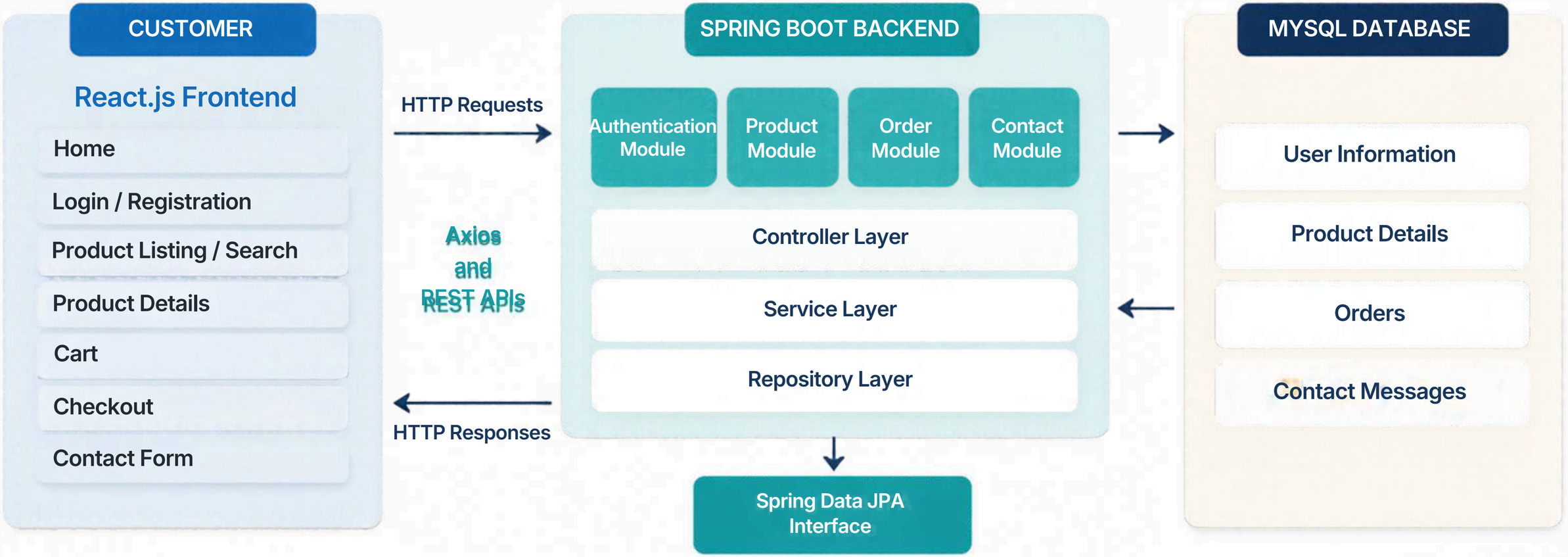

Git


GitHub

TECHNOLOGY ROLES

- React.js builds interactive UI
- Spring Boot manages backend logic and REST APIs
- Spring Security and BCrypt handle authentication and password encryption
- Spring Data JPA simplifies DB operations
- MySQL stores application data
- Axios connects frontend to backend APIs

System Architecture



NOTE

Browser localStorage maintains shopping cart data.

Key Modules Implemented



User Authentication

Registration & login

Input validation, BCrypt password encryption

Authentication checks before order placement



Product Management

Product listing and details

Browsing and search functionality



Shopping Cart

Add/remove products

Update quantities

Cart data stored in browser localStorage



Order Management

Checkout and order submission

Backend validation of authenticated users

Store order and order-item data



Contact Management

Customer inquiry form

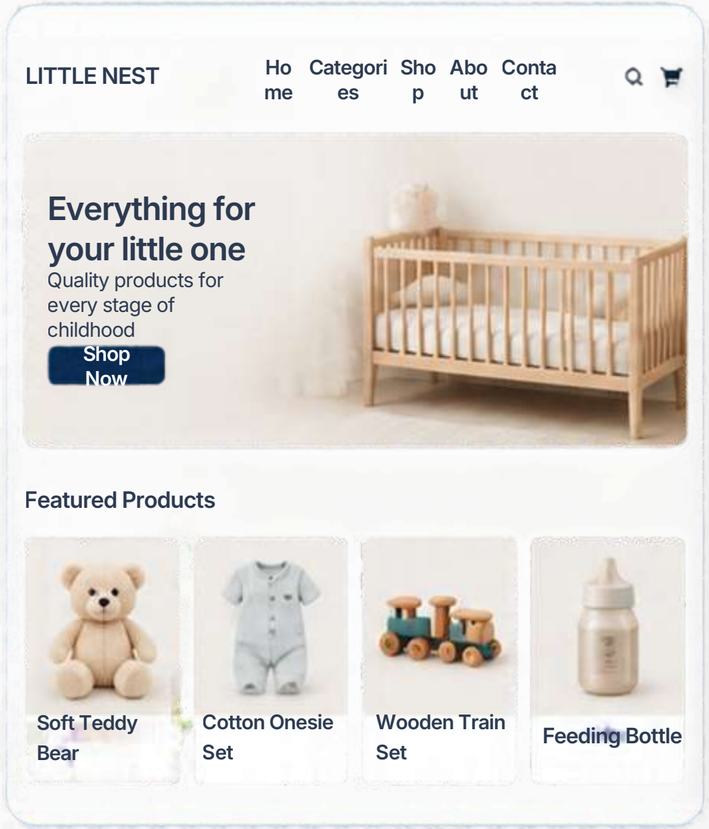
Contact messages stored in MySQL

1 Browse Products → 2 Add to Cart → 3 Checkout & Place Order → 4 Order Confirmed ⋯ 5 Get in Touch

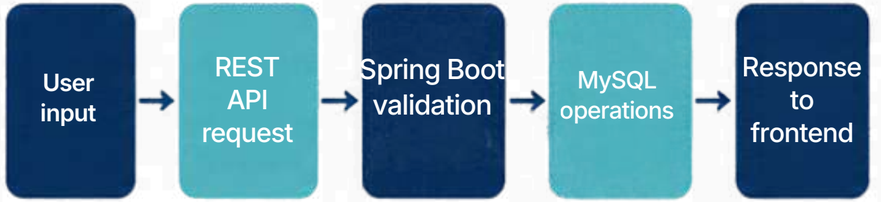
Application Workflow and Implementation

CUSTOMER WORKFLOW

1. User opens website
2. Browses/searches products
3. Selects product, adds to cart
4. Registers or logs in
5. Reviews cart, proceeds to checkout
6. Backend validates authentication, processes order
7. Order details saved in MySQL
8. Order confirmation returned



AUTHENTICATION WORKFLOW



TECHNOLOGIES SUPPORTING THE WORKFLOW

- React Router for navigation
- Axios for API requests
- Spring Data JPA for database access

Payment gateway integration is future work.

Testing and Internship Outcomes

PROJECT TESTING



- User registration & input validation
- Login validation & invalid credential handling
- Product listing & detail retrieval
- Search & cart operations
- Checkout & order creation
- Contact form submission
- Frontend-backend integration testing

REPORTED OUTCOMES

Core modules tested with validation for duplicates, invalid inputs, unauthorized requests

Integration of React, Spring Boot, MySQL via REST APIs

Responsive, user-friendly interface

SKILLS GAINED

- **React** — **React** frontend development
- **Java** — **Java & Spring Boot** backend development
- **API** — **REST API** integration
- **MySQL** — **MySQL** database management
- **Auth** — **Authentication &** password encryption
- **Dev** — **Debugging, integration testing,** documentation

Conclusion and Future Scope

CONCLUSION

Demonstrated practical full-stack development integrating React.js frontend, Spring Boot backend, REST APIs, Spring Data JPA, and MySQL

Implemented core e-commerce features: authentication, product browsing, cart management, order creation, customer contact

Strengthened practical knowledge of full-stack web development and technology integration

FUTURE ENHANCEMENTS

Secure online payment gateway integration

JWT-based authentication and role-based authorization

Order tracking

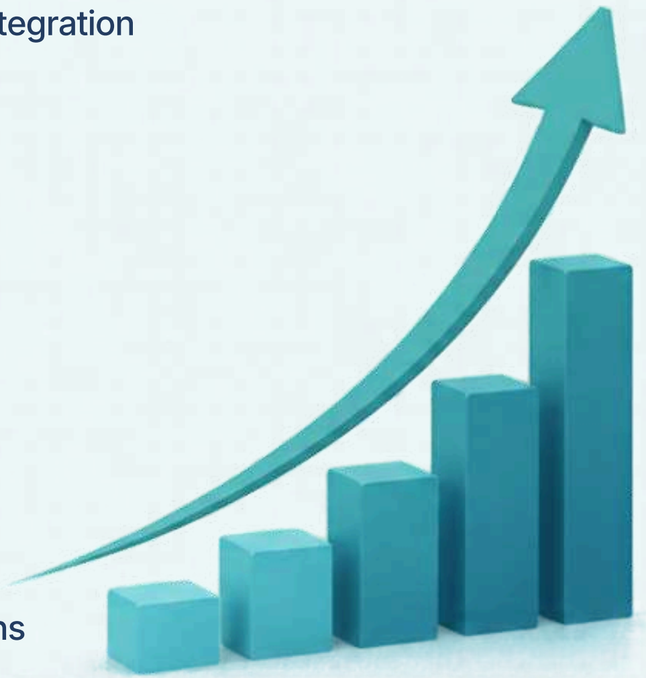
Email and SMS notifications

Inventory management

Product ratings and reviews

Cloud deployment

AI-based product recommendations



Thank You